Quantification: 542_z06
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 19 |
| interior lacuna count | 15 |
| frame-edge lacuna count | 4 |

Interior lacunae only: mean, median and sample SD (n = 15)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2576 | 3013 | 1380.42 | px^2 |
| major_axis_length_px | lacuna shape | 109.567 | 124.464 | 38.2072 | px |
| minor_axis_length_px | lacuna shape | 31.8891 | 33.1799 | 10.1914 | px |
| aspect_ratio | lacuna shape | 3.6142 | 3.6285 | 1.2579 | unitless |
| eccentricity | lacuna shape | 0.9409 | 0.9613 | 0.0498 | unitless |
| solidity | lacuna shape | 0.7493 | 0.7672 | 0.1286 | unitless |
| roots_count | headline | 14.0667 | 16 | 6.4195 | count |
| ring_length_r30_px | headline | 447.667 | 410 | 188.736 | px |
| ring_length_r60_px | reported | 1016.07 | 960 | 429.438 | px |
| owned_length_px | ownership-dependent | 2004.95 | 2091.63 | 1454.79 | px |
| edge_count | ownership-dependent | 88.7333 | 75 | 65.1662 | count |
| mean_edge_length_px | ownership-dependent | 22.4129 | 22.7351 | 3.7601 | px |
| perimeter_px | normalised | 292.07 | 309.522 | 122.808 | px |
| ring_area_r30_px2 | normalised | 10491.5 | 11337 | 3304.03 | px^2 |
| ring_area_r60_px2 | normalised | 24400.9 | 26961 | 7815.16 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0.0001 | unitless |
| in_frame_fraction_r60 | normalised | 0.9858 | 1 | 0.0377 | unitless |
| ring_density_r30 | normalised | 0.0417675 | 0.041882 | 0.00694049 | px^-1 |
| ring_density_r60 | normalised | 0.0405016 | 0.0397432 | 0.00624729 | px^-1 |
| roots_per_100px_perimeter | normalised | 5.0246 | 4.7443 | 1.3785 | per 100 px |
| ring_attached_length_r30_px | network v2 | 392.867 | 351 | 192.46 | px |
| ring_attached_length_r60_px | network v2 | 810.2 | 691 | 449.189 | px |
| ring_length_w_r30_px | network v2 | 509.124 | 464.262 | 216.705 | px |
| ring_length_w_r60_px | network v2 | 1153.17 | 1075.26 | 492.452 | px |
| sholl_crossings_r10 | network v2 | 9.6 | 10 | 4.8077 | count |
| sholl_crossings_r20 | network v2 | 13.2667 | 14 | 5.3515 | count |
| sholl_crossings_r30 | network v2 | 13.8667 | 13 | 5.2762 | count |
| ring_width_mean_r30_px | width | 7.5443 | 7.5189 | 0.4181 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.0064e+06 | px^2 |
| total_skeleton_length_px | 39229 | px |
| canalicular_length_density_per_px | 0.0389794 | px^-1 |
| junction_count | 869 | count |
| junction_density_per_px2 | 0.00086347 | px^-2 |
| lacuna_count | 19 | count |
| lacunae_per_px2 | 1.888e-05 | px^-2 |
| skeleton_component_count | 250 | count |

| | | |
|---|---|---|
| mean_component_length_px | 156.916 | px |
| median_component_length_px | 26 | px |
| field_length_density_w_per_px | 0.0441542 | px^-1 |
| field_density_without_flagged_per_px | 0.0389281 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 375 | px |
| unattached_component_count | 20 | count |
| canalicular_length_density_attached_per_px | 0.0389794 | px^-1 |
| field_length_density_with_unattached_per_px | 0.039352 | px^-1 |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 8.1584 | px |
| width_median_px | 8.2462 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.7703 | px |
| width_pixels_used | 37983 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1381 | 54.5303 | 43.8441 | 1.2437 | 0.5946 | 0.657 | -0.6304 | 16.7227 | 909.274 | True |
| 2 | 445 | 56.9837 | 11.5357 | 4.9398 | 0.9793 | 0.7672 | -0.2356 | 148.654 | 554.791 | False |
| 3 | 1001 | 62.1533 | 28.1242 | 2.21 | 0.8918 | 0.5533 | -0.1968 | 55.7832 | 107.949 | False |
| 4 | 895 | 46.7108 | 27.3564 | 1.7075 | 0.8106 | 0.7956 | 0.2016 | 143.341 | 635.06 | False |
| 5 | 2450 | 125.59 | 27.3191 | 4.5971 | 0.9761 | 0.6618 | -0.1995 | 179.65 | 380.673 | False |
| 6 | 4645 | 128.217 | 49.4579 | 2.5925 | 0.9226 | 0.72 | -0.0979 | 201.844 | 870.357 | False |
| 7 | 2177 | 80.2406 | 35.2894 | 2.2738 | 0.8981 | 0.8959 | -0.0663 | 304.003 | 624.629 | False |
| 8 | 3251 | 124.464 | 40.9538 | 3.0391 | 0.9443 | 0.639 | 0.028 | 423.426 | 66.5638 | False |
| 9 | 3239 | 125.767 | 33.4856 | 3.7559 | 0.9639 | 0.9114 | -0.0886 | 481.731 | 573.277 | False |
| 10 | 3188 | 122.408 | 33.7355 | 3.6285 | 0.9613 | 0.8804 | -0.0452 | 529.624 | 391.03 | False |
| 11 | 3013 | 170.645 | 29.6128 | 5.7626 | 0.9848 | 0.5248 | -0.2083 | 594.958 | 912.812 | False |
| 12 | 3414 | 148.999 | 33.1799 | 4.4906 | 0.9749 | 0.8294 | -0.0805 | 580.581 | 782.628 | False |
| 13 | 4647 | 139.214 | 45.6902 | 3.0469 | 0.9446 | 0.8623 | -0.0076 | 723.033 | 247.267 | False |
| 14 | 1123 | 92.9757 | 18.8182 | 4.9407 | 0.9793 | 0.6997 | -0.1287 | 817.288 | 724.433 | False |
| 15 | 3750 | 145.258 | 33.093 | 4.3894 | 0.9737 | 0.8965 | -0.0485 | 952.204 | 400.276 | True |
| 16 | 1341 | 90.1135 | 19.5596 | 4.6071 | 0.9762 | 0.911 | -0.0956 | 984.752 | 948.262 | True |
| 17 | | | | | | | | 310.754 | 226.539 | False |
| 18 | | | | | | | | 933.644 | 3.9103 | True |
| 19 | | | | | | | | 368.378 | 576.748 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 218 | 581 | 894.411 | 31 | 28.852 | 7.8557 |
| 2 | 6 | 194 | 502 | 825.027 | 30 | 27.5009 | 7.7664 |
| 3 | 13 | 368 | 687 | 1456.15 | 53 | 27.4746 | 7.6771 |
| 4 | 9 | 255 | 657 | 344.522 | 19 | 18.1327 | 7.3342 |
| 5 | 10 | 380 | 1031 | 2222.31 | 97 | 22.9104 | 7.2803 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 27 | 784 | 1661 | 5469.89 | 222 | 24.6392 | 7.6259 |
| 7 | 16 | 410 | 749 | 822.198 | 36 | 22.8388 | 7.4551 |
| 8 | 17 | 459 | 883 | 919.968 | 44 | 20.9084 | 7.5163 |
| 9 | 7 | 388 | 960 | 2091.63 | 92 | 22.7351 | 8.6437 |
| 10 | 17 | 465 | 1270 | 3280.28 | 134 | 24.4797 | 7.7735 |
| 11 | 20 | 767 | 1786 | 4088.09 | 215 | 19.0144 | 7.4169 |
| 12 | 19 | 601 | 1342 | 2217.9 | 106 | 20.9236 | 7.1869 |
| 13 | 20 | 608 | 1318 | 2788.43 | 136 | 20.5032 | 7.6692 |
| 14 | 9 | 318 | 855 | 2153.65 | 75 | 28.7154 | 6.6409 |
| 15 | 6 | 274 | 679 | 862.997 | 44 | 19.6136 | 7.6559 |
| 16 | 5 | 148 | 499 | 95.4264 | 7 | 13.6323 | 5.8635 |
| 17 | 17 | 564 | 1304 | 1184.29 | 58 | 20.4188 | 7.6596 |
| 18 | 2 | 69 | 189 | 389.948 | 18 | 21.6638 | 8.4397 |
| 19 | 4 | 154 | 236 | 209.983 | 14 | 14.9988 | 7.5189 |

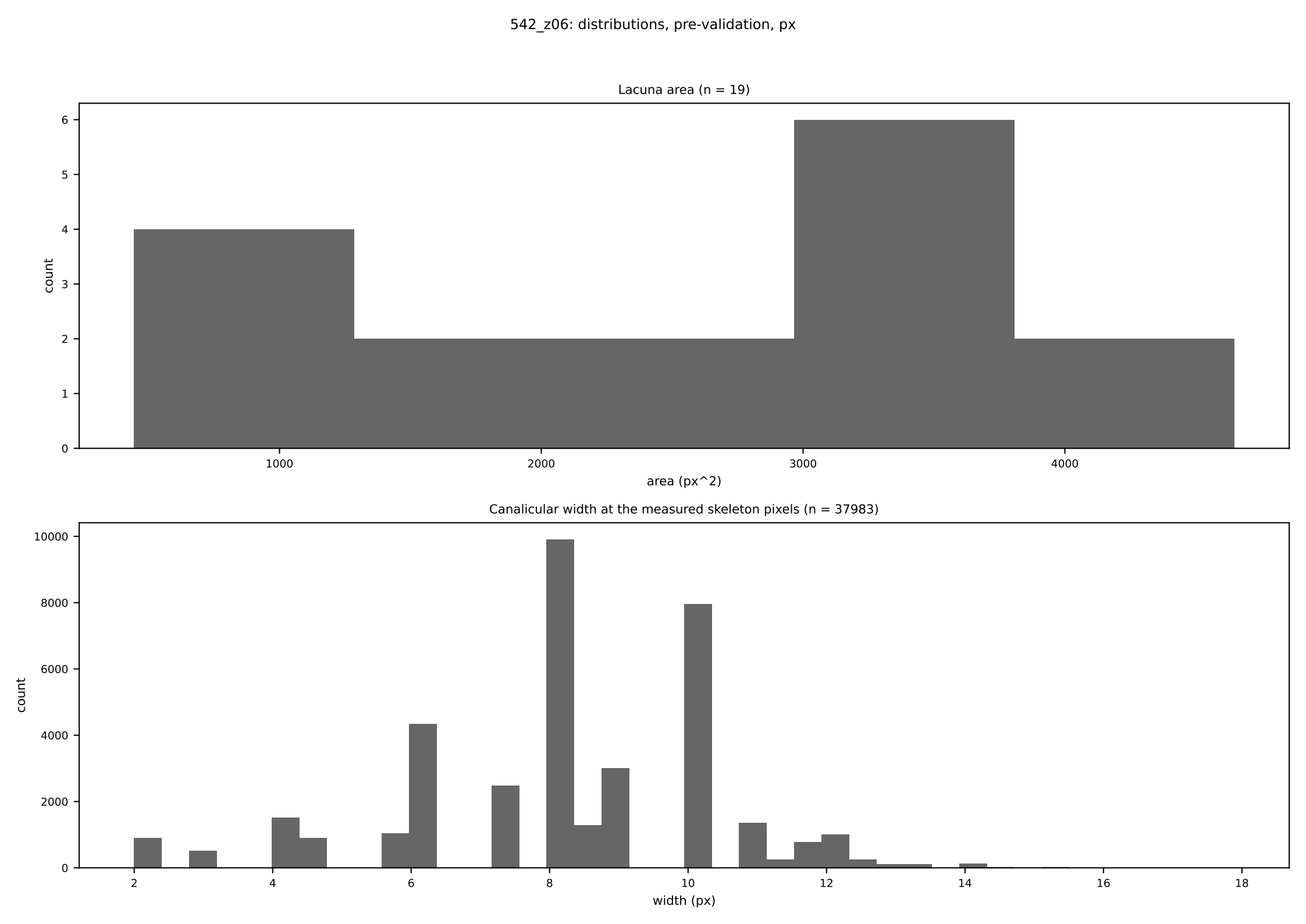
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 215.232 | 5617 | 12694 | 0.6363 | 0.5983 | 0.0388107 | 0.0457697 | 3.2523 |
| 2 | 126.468 | 6226 | 14794 | 1 | 1 | 0.0311597 | 0.0339327 | 4.7443 |
| 3 | 225.309 | 8379 | 19327 | 0.9998 | 0.884 | 0.0439193 | 0.0355461 | 5.7699 |
| 4 | 151.782 | 6752 | 15321 | 1 | 1 | 0.0377666 | 0.0428823 | 5.9296 |
| 5 | 322.043 | 11663 | 28685 | 1 | 1 | 0.0325817 | 0.0359421 | 3.1052 |
| 6 | 438.517 | 13901 | 31616 | 1 | 1 | 0.0563988 | 0.0525367 | 6.1571 |
| 7 | 211.924 | 8423 | 18846 | 1 | 1 | 0.0486762 | 0.0397432 | 7.5499 |
| 8 | 359.191 | 12383 | 26961 | 1 | 0.9025 | 0.037067 | 0.032751 | 4.7329 |
| 9 | 302.451 | 10940 | 24775 | 1 | 1 | 0.0354662 | 0.0387487 | 2.3144 |
| 10 | 309.522 | 11337 | 28286 | 1 | 1 | 0.0410161 | 0.0448985 | 5.4923 |
| 11 | 468.274 | 15363 | 34096 | 1 | 1 | 0.0499251 | 0.0523815 | 4.271 |
| 12 | 422.978 | 13165 | 30001 | 1 | 1 | 0.0456514 | 0.0447318 | 4.492 |
| 13 | 379.848 | 13296 | 32108 | 1 | 1 | 0.045728 | 0.041049 | 5.2653 |
| 14 | 217.924 | 8830 | 22986 | 1 | 1 | 0.0360136 | 0.0371966 | 4.1299 |
| 15 | 332.894 | 10505 | 23806 | 0.8602 | 0.7919 | 0.0260828 | 0.0285222 | 1.8024 |
| 16 | 205.776 | 6766 | 16357 | 0.7841 | 0.714 | 0.0218741 | 0.0305068 | 2.4298 |
| 17 | 387.948 | 13037 | 31120 | 1 | 1 | 0.0432615 | 0.0419023 | 4.382 |
| 18 | 87.0061 | 2915 | 8653 | 0.5433 | 0.5282 | 0.0236707 | 0.0218421 | 2.2987 |
| 19 | 56.8701 | 3677 | 7091 | 1 | 1 | 0.041882 | 0.0332816 | 7.0336 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 218 | 574 | 251.066 | 663.701 | 6 | 5 | 5 |
| 2 | 157 | 376 | 224.137 | 565.517 | 4 | 8 | 10 |
| 3 | 308 | 632 | 423.99 | 782.24 | 7 | 7 | 10 |
| 4 | 211 | 338 | 285.966 | 736.257 | 5 | 7 | 8 |
| 5 | 319 | 837 | 437.647 | 1179.53 | 8 | 14 | 13 |
| 6 | 781 | 1566 | 902.465 | 1908.83 | 20 | 20 | 22 |
| 7 | 351 | 547 | 464.262 | 849.512 | 8 | 12 | 11 |
| 8 | 415 | 691 | 514.304 | 987.423 | 10 | 11 | 10 |
| 9 | 301 | 537 | 436.09 | 1075.26 | 11 | 14 | 12 |
| 10 | 444 | 1094 | 534.517 | 1452.87 | 10 | 14 | 15 |
| 11 | 686 | 1687 | 873.654 | 2015.19 | 14 | 22 | 25 |
| 12 | 530 | 1121 | 674.286 | 1515.8 | 14 | 16 | 15 |
| 13 | 570 | 1215 | 688.671 | 1491.87 | 15 | 22 | 19 |
| 14 | 227 | 546 | 358.907 | 969.495 | 5 | 11 | 14 |
| 15 | 187 | 280 | 305.522 | 756.813 | 7 | 9 | 14 |
| 16 | 113 | 113 | 164.368 | 551.961 | 4 | 7 | 6 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 17 | 485 | 842 | 650.156 | 1509.59 | 11 | 16 | 18 |
| 18 | 36 | 97 | 79.7696 | 213.267 | 1 | 3 | 4 |
| 19 | 108 | 124 | 167.811 | 258.166 | 2 | 5 | 6 |


542_z06: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
1
2
3
4
5
6
1000
2000
3000
4000
Canalicular width at the measured skeleton pixels (n = 37983)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18